

SAI DHARANI GARIKAPATI

Full Stack Developer

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SUMMARY

Versatile Python Full Stack & AI Engineer with 4+ years of experience developing scalable web, cloud, and AI applications across enterprise and startup environments. Skilled in FastAPI, React, AWS, PyTorch, and LangChain, integrating full-stack engineering with LLM-based automation, data pipelines, and MLOps deployments. Proven ability to build secure, high-performance systems, implement CI/CD automation, and deliver production-grade Generative AI and analytics platforms.

EXPERIENCE

Full Stack Software Engineer | Perplexity AI, USA

Feb 2024 – Present

- Architected and deployed end-to-end AI web applications using Python (FastAPI, Flask) and React.js, building RESTful APIs, authentication layers, and real-time dashboards for analytics products.
- Developed scalable microservices architecture supporting over 100K daily API calls with async I/O, Redis caching, and PostgreSQL/MySQL backends optimized using stored procedures and connection pooling.
- Built and maintained responsive front-end components using React, TypeScript, and Tailwind CSS, integrating backend APIs, JWT authentication, and reusable hooks to enhance UX performance by 30%.
- Designed automated ETL ingestion pipelines leveraging Pandas, SQLAlchemy, and Airflow, orchestrating data cleaning, validation, and transformation workflows across AWS S3 buckets and multiple API sources.
- Integrated OpenAI and LangChain APIs for intelligent query handling, embedding Python-based LLMs into production to generate summarized insights and personalized recommendations in real time.
- Containerized and deployed full-stack environments using Docker and Kubernetes, configuring auto-scaling pods and load balancing on AWS EKS and EC2 instances with Nginx reverse proxy.
- Implemented end-to-end CI/CD pipelines through GitHub Actions and Jenkins, automating testing, code linting, and deployments across staging and production environments seamlessly and efficiently.
- Enhanced observability and monitoring using ELK Stack (Elasticsearch, Logstash, Kibana) and Prometheus/Grafana, providing real-time metrics and alerts for latency, uptime, and API health.
- Collaborated with data science teams to deploy PyTorch and TensorFlow-based machine learning micro-APIs, integrating model inference endpoints into FastAPI for real-time production usage.
- Optimized application security and compliance using OAuth 2.0, RBAC, parameterized SQL queries, and TLS-encrypted data transfers with AWS KMS for enterprise-grade protection.

Full Stack Engineer | Accenture, Bengaluru, India

Jan 2020 – Jul 2023

- Led development of Python FastAPI and Flask microservices integrated with React and PostgreSQL, powering enterprise-grade dashboards and cutting API response times by over 40%.
- Built end-to-end data pipelines using PySpark, Pandas, and Airflow on AWS (S3, Glue, Lambda), automating ETL workflows for 50M+ records across multiple business units.
- Architected and deployed full-stack web applications with React, Redux, REST APIs, and GraphQL, improving scalability, modularity, and user engagement for high-traffic enterprise platforms.
- Designed optimized PostgreSQL and MongoDB data models and implemented SQLAlchemy ORM queries, reducing query latency by 30% while supporting multi-tenant microservice architecture.
- Integrated machine learning models using TensorFlow and Scikit-learn into backend APIs for anomaly detection and real-time predictions, improving model accuracy by over 25%.
- Automated DevOps workflows using Docker, Kubernetes, and GitHub Actions, enabling zero-downtime CI/CD releases and reducing deployment cycles from weekly to daily operations.
- Developed serverless cloud functions and APIs using AWS Lambda, DynamoDB, API Gateway, and Terraform, reducing infrastructure costs by 35% across multiple client environments.
- Implemented robust authentication and access control using OAuth2, JWT, and RBAC, ensuring SOC2-compliant security standards across multi-environment enterprise deployments.
- Created interactive visualization dashboards using Plotly, D3.js, and Power BI, transforming raw datasets into actionable insights for finance, supply chain, and marketing stakeholders.
- Partnered with Agile cross-functional teams (UI/UX, Data Engineering, and MLOps) to deliver production-grade analytics and AI platforms, improving project delivery velocity by 45%.

TECHNICAL SKILLS

Languages & Programming: Python, SQL, Java, C, Bash, JavaScript (ES6+), TypeScript, HTML5, CSS3

Machine Learning & Generative AI: PyTorch, TensorFlow, Scikit-learn, XGBoost, LightGBM, Hugging Face Transformers, LLaMA (2/3), GPT, BERT, CLIP, Diffusion Models (Emu, EmuEdit, EmuVideo), Reinforcement Learning (RL, RLHF), Retrieval-Augmented Generation (RAG), Prompt Engineering, Recommendation Systems (GEM, Andromeda), Generative Ad Copy, Conversational AI

Natural Language Processing (NLP): spaCy, NLTK, Named Entity Recognition (NER), Sentiment Analysis, Document Parsing, OCR (Tesseract, AWS Textract)

MLOps & Model Deployment: MLflow, Kubeflow, TorchServe, Triton Inference Server, ONNX, TorchScript, NVIDIA NIM, TensorRT, Model Quantization (FP8/INT4), CI/CD (Jenkins, GitHub Actions), A/B Testing, Canary & Shadow Deployments, Model Monitoring Dashboards

DevOps & Containerization: Docker, Kubernetes, Airflow, DVC, Git (GitHub, GitLab, Bitbucket), Jenkins, Terraform, Agile/Scrum

Data Engineering & Big Data: Apache Spark (PySpark, Spark SQL), Kafka, Flink, Presto, Hive, Scuba, ETL Pipelines, Distributed Systems, Scalable Data Pipelines, Pandas, NumPy

Cloud & Infrastructure: AWS (SageMaker, Lambda, S3, Glue, EKS, IAM, KMS, CloudWatch), Azure (ML Studio, Databricks), GCP (Vertex AI), Meta MTIA, Elasticsearch, Cloud-Native ML Architectures

Vector Databases & RAG Systems: Pinecone, FAISS, NeMo Retriever, Redis Feature Store

Responsible AI & Governance: SHAP, LIME, Fairlearn, Bias Detection, AI Watermarking, Brand Safety Filters, Human-in-the-Loop Systems, Responsible AI Practices, GDPR, KYC/AML Compliance

APIs & System Integration: FastAPI, Flask, Django, RESTful APIs, GraphQL, ERP Integrations

Monitoring & Observability: Prometheus, Grafana, ELK Stack (Elasticsearch, Logstash, Kibana), OpenTelemetry

Visualization & Analytics: Plotly, D3.js, Power BI, Tableau

Soft Skills / Collaboration: Cross-functional Collaboration, Stakeholder Communication, AI Safety & Compliance, Technical Documentation, Knowledge Sharing, Problem Solving, Leadership in Agile Environments

EDUCATION

Master of Science – Computer Science

Texas Tech University, Lubbock, TX

Bachelor of Technology – Computer Science and Engineering

DVR & Dr. HS MIC College of Technology, India